Chapter 1

Speech Communication & Group Working Skills

Learning objectives

- Define speech communication and describe its importance in technical contexts.
- Identify engineers' responsibilities and relate them to professional communication.
- Comprehend the group discussion.
- Understand the structure and flow of academic debates.
- Organize a group project using tools like Gantt Charts and milestones.

Learning objectives (Cont.)

- Apply structured problem-solving frameworks.
- Identify common interpersonal conflicts in teams.
- Strategies for conflict resolution.
- Collaborate in groups to analyze and solve real-world scenarios.

1. Speech Communication

What is Communication?

- Communication is the process of sharing information, thoughts, and feelings between people through speaking, writing, or body language.
- Types of Communication
  - *Spoken or verbal communication*: face-to-face, telephone, radio or television and other media.
  - *Non-verbal communication*: body language, gestures, how we dress or act, where we stand, and even our scent.
    - For example, the tone of voice can give clues to mood or emotional state, whilst hand signals or gestures can add to a spoken message.

Source: https://www.skillsyouneed.com/ips/what-is-communication.html

What is Communication? (Cont.)

- Communication is the process of sharing information, thoughts, and feelings between people through speaking, writing, or body language.

- Types of Communication

  - *Written communication*: letters, e-mails, social media, books, magazines, the Internet, and other media.

  - *Visualizations*: graphs and charts, maps, logos and other visualizations can all communicate messages.

The Communication Process

- A message or communication is sent by the sender through a communication channel to a receiver, or to multiple receivers.
  - ❖ The sender must encode the message (the information being conveyed) into a form that is appropriate to the communication channel, and the receiver(s) then decodes the message to understand its meaning and significance.

The Communication Process (Cont.)

- Misunderstanding can occur at any stage of the communication process.
  - ❖ Effective communication involves minimising potential misunderstanding and overcoming any barriers to communication at each stage in the communication process.

The Communication Process (Cont.)

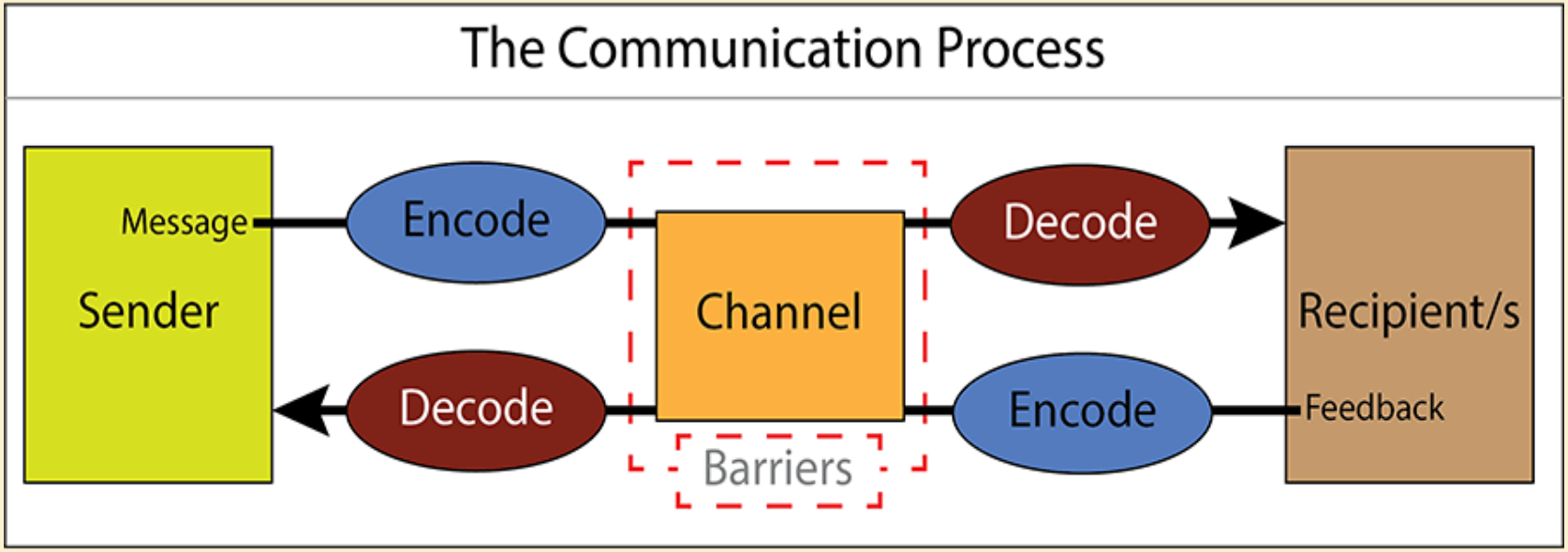


Figure 1: The Communication Process. Source: https://www.skillsyouneed.com/ips/what-is-communication.html

The Communication Process (Cont.)

- An effective communicator understands their audience, sender chooses an appropriate communication channel, hones his/her message to this channel and encodes the message to reduce misunderstanding by the receiver(s).
- Sender will also seek out feedback from the receiver(s) as to how the message is understood and attempt to correct any misunderstanding or confusion as soon as possible.
- Receiver(s) can use techniques such as *Clarification* and *Reflection* as effective ways to ensure that the message sent has been understood correctly.

Importance in Technical Careers

- Engineers work in cross-functional teams.
- Need to present technical data clearly to diverse audiences.
- Involved in meetings, project documentation, proposals.
- Effective communication boosts project success, safety, and collaboration.

Engineers' Responsibilities & Professional Ethics

- Ethical communication: honesty, transparency, and accountability.
- Professional conduct: respecting confidentiality, avoiding plagiarism.
- Responsibilities: reporting truthfully, public safety, and teamwork.
- Communication is essential for ethical decision-making.

Soft Skills for Engineers

- Public speaking and presentation skills.
- Interpersonal communication.
- Team collaboration.
- Conflict resolution.
- Emphasis on clear, concise, and respectful exchanges.

Barriers to Effective Communication

- Physical: noise, distance, technical problems.
- Psychological: stress, fear, prejudice.
- Language: jargon, unfamiliar accents.
- Cultural: differences in norms, values, expectations.
- Others.

→ *How can these be minimized in technical workplaces?*

In conclusion

- What is Communication?
- The Communication Process.
- Importance in Technical Careers.
- Engineers' Responsibilities & Professional Ethics.
- Soft Skills for Engineers.
- Barriers to Effective Communication.

2. Group Working Skills

Group Discussion

Introduction to Group Discussion

- Group discussion is a structured conversation among participants to share ideas, analyze issues, and reach consensus.
- Purpose of group discussion evaluates communication, teamwork, critical thinking, and interpersonal skills.
- Real-world relevance: Common in academic, professional, and corporate settings.

Introduction to Group Discussion (Cont.)

- Objectives of group discussion
  - ❖ Generate multiple solutions through collaboration.
  - ❖ Encourage idea sharing and critical reflection.
  - ❖ Reveal hidden knowledge and personality traits.
  - ❖ Enhance knowledge base and foster consensus-building.

Employers' Expectations

- Initiative and proactive behavior.
- Clarity of thought and logical expression.
- Team-building and collaboration.
- Relevant content and knowledge depth.
- Emotional intelligence and maturity.
- Leadership and interpersonal skills.
- Time management and presentation skills.

Pre-Session Preparation

- Read books, articles, and websites about group discussion.
- Self-assess strengths and weaknesses.
- Conduct mock group discussions with peers.
- Practice converting informal talks into structured discussions.

Rules and Norms

- Ideal group size: 5 to 10 participants.
- Time duration: 10–15 minutes.
- Seating formation: circular or rectangular.
- Ensure all participants get a chance to speak.

Role of the Moderator

- Ensure focus and manage conflict.
- Maintain decorum and encourage participation.
- Offer unbiased and constructive feedback.
- Stimulate discussion and synergy.

Tips During Discussion

- Speak clearly and confidently.
- Focus on the topic, not the moderator.
- Respect others' time and opinions.
- Support arguments with data or stories.
- Show leadership through inclusion and synthesis.
- Display positive body language.

Post-Discussion Feedback

- Assessment areas
  - ❖ Attitude (e.g., liveliness, openness).
  - ❖ Confidence (e.g., participation, interest).
  - ❖ Communication (e.g., politeness, fluency).
  - ❖ Interpersonal skills (e.g., adaptability, impact).
  - ❖ Flow of thoughts (e.g., logic, originality).
- Feedback scale
  - ❖ 1 – Poor; 2 – Below Average; 3 – Average; 4 – Good; 5 – Excellent.

Post-Session Reflection & Takeaways

- Self-evaluate performance: What worked? What didn't?
- Practice regularly to build confidence.
- Group discussion participation = learning + networking + growth.
- Take group discussions as opportunities to become a complete corporate personality.

Some Patterns For Discussion

- ***Some patterns for starting a discussion***
  - We have assembled here to discuss…
  - Let us get down to business…
  - Shall we set the ball rolling?
  - Shall we make a start?
  - Etc.

Some Patterns For Discussion (Cont.)

- ***Some patterns for interrupting a discussion***
  - ❖ Sorry to interrupt you…
  - ❖ Excuse me, but…
  - ❖ Could I make a suggestion, please?
  - ❖ Could I say something…?
  - ❖ Sorry to disagree with you.
  - ❖ If I could make a point here…
  - ❖ Etc.

Some Patterns For Discussion (Cont.)

- ***Some patterns for ending a discussion***
  - ❖ I think that covers everything.
  - ❖ It is time to wind up.
  - ❖ Shall we close the discussion then?
  - ❖ Etc.

Practice 1

Group Discussion

Topic: What communication challenges do engineers face in their workplaces or internships?

- ***Your tasks:***
  - 5 - 10 members per group.
  - Discuss for 15 minutes.
  - Present your group's top 2 challenges and explain why.
  - The classmates vote on clarity, logic, and delivery.

In conclusion

- Introduction to Group Discussion
- Employers' Expectations
- Pre-Session Preparation
- Rules and Norms
- Role of the Moderator
- Tips During Discussion
- Post-Discussion Feedback
- Post-Session Reflection & Takeaways
- Some Patterns For Discussion

Debate &
Working Planning Skills

What is Debate?

- Debate is a structured argument involving two opposing sides on a given topic.
- It enhances:
  - ❖ Critical thinking.
  - ❖ Persuasive communication.
  - ❖ Evidence-based reasoning.
  - ❖ Respectful disagreement.

Structure of a Debate

- *Motion/Resolution*: A clear statement to argue for or against.
- *Affirmative team*: supports the motion.
- *Negative team*: opposes the motion.
- *Each team delivers*:
  1. Opening statement.
  2. Arguments with evidence.
  3. Rebuttals to opposition.
  4. Conclusion: Summary + final appeal.
- *Example Motion*: "*Nuclear energy should replace fossil fuels globally*."

Building an Argument (Claim – Evidence – Reasoning)

- *Claim:* Your position or idea.
- *Evidence:* Data, examples, expert opinion.
- *Reasoning:* Explain how the evidence supports your claim.

❑ ***Example***

- *Claim*: Nuclear energy is low in carbon emissions.
- *Evidence*: According to IEA (2022), nuclear plants emit <50g CO_2/kWh.
- *Reasoning*: This makes nuclear power a cleaner long-term energy source.

Rebuttal Techniques

- *Refutation*: Directly disproving a point.
- *Counterargument*: Presenting an alternative view.
- *Concession*: Agreeing with part of the argument but shifting focus.
- Use polite, logical language:
  - ❖ “However, that doesn’t consider...”
  - ❖ “This overlooks the fact that...”
  - ❖ “We agree, but our concern is...”

Language for Debate

Purposes	Examples
Stating opinion	I strongly believe...; It is clear that…
Disagreeing	I see your point, but…; However…
Supporting ideas	According to…; Research shows…
Rebutting	This may seem true, but…; On the contrary…

Reflection Debrief

- *After the debate*
  - ❖ What made some arguments more convincing than others?
  - ❖ How did rebuttals impact the outcome?
  - ❖ What would you improve in your next debate?

Practice 2

Debate

Topic options (selected or assigned):

1. AI will eliminate more jobs than it creates.
2. Online education is more effective than in-person learning.
3. Social media does more harm than good.
4. Engineers should learn more about public speaking.

- ***Your tasks:***
  - 5 - 10 members per group.
  - 15 minutes preparation.
  - Separated into two sides for presentation: 2-minute argument + 1-minute rebuttal.
  - The classmates vote on clarity, logic, and delivery.

Why is planning important in group work?

- Prevents last-minute rush.
- Clarifies tasks, roles, and deadlines.
- Helps monitor progress and manage workload.
- Enhances collaboration and accountability.

Gantt Chart: What & Why?

- Gantt Chart = visual project timeline.
- Shows tasks, durations, dependencies.
- Ideal for collaborative assignments (e.g., presentations).
- Tools: Google Sheets, MS Excel, MS Project, etc.
- Example of Gantt Chart and milestones.

Tasks	Week 1	Week 2	Week 3	Week 4
Topic selection	Δ			
Research content	Δ	Δ		
Slide creation		Δ	Δ	
Rehearsal			Δ	Δ
Final delivery				Δ

Practice 3

Group Planning Activity

- Choose a tentative presentation topic.
- Identify 4–5 main tasks.
- Draft a simple Gantt chart (on paper or whiteboard).
- Assign initial roles.

- ***Your tasks:***
  - 5 - 10 members per group.
  - 15 minutes preparation.
  - Topic Presentation.
  - The classmates vote on clarity, logic, and delivery.

In conclusion

- What is Debate?
- Structure of a Debate.
- Building an Argument.
- Rebuttal Techniques.
- Language for Debate.
- Reflection Debrief.
- Why is planning important in group work?
- Gantt Chart: What & Why?

Solving problems

What is a Problem?

- "A problem is the gap between a current situation and a desired outcome."
- In group settings, problems may involve:
  - ❖ Miscommunication.
  - ❖ Missed deadlines.
  - ❖ Role ambiguity.
  - ❖ Unequal contribution.

Structured Problem-Solving Frameworks?

Method	Description	Use When
Root Cause Analysis (5 Whys)	Repeatedly asking “Why?” to find underlying cause	For recurring or unclear problems
Brainstorming	Unstructured idea generation	To find multiple solutions
SWOT Analysis	Strengths, Weaknesses, Opportunities, Threats	To evaluate team options
Fishbone Diagram	Cause-effect mapping	For visualizing complex issues

The 5 Whys Technique - Example

- **Problem**: Group submitted the presentation late.
  - ❖ Why? → We didn't finish on time.
  - ❖ Why? → One member didn’t submit slides.
  - ❖ Why? → They misunderstood their task.
  - ❖ Why? → The role was not clearly assigned.
  - ❖ Why? → No planning in the initial meeting.

➡ **Root Cause**: Lack of initial role clarification.

Brainstorming Tips

- No criticism during idea generation.
- Encourage quantity over quality (at first).
- Build on others' ideas ("Yes, and…").
- Use sticky notes, whiteboards, or mind maps.

- **Quick brainstorm**
  - "How can we reduce conflict during online teamwork?"

Common Sources of Conflict in Teams

- *Task conflict:* disagreement on what to do.
- *Process conflict:* disagreement on how to do it.
- *Relationship conflict:* interpersonal tension.
- *Role conflict:* unclear or overlapping responsibilities.
- *Status conflict:* dominance, competition, ego.

Conflict Escalation Model

- Visual showing stages of conflict
  - ❖ Discomfort.
  - ❖ Incident.
  - ❖ Misunderstanding.
  - ❖ Tension.
  - ❖ Crisis.

➡ At which stage should we intervene?

Conflict Resolution Strategies

Strategy	When to Use	Example Phrase
Active Listening	Always	“So you’re saying that...”
I-Messages	To express concern	“I feel frustrated when…”
Compromise	Moderate disagreement	“Let’s meet halfway…”
Collaboration	For long-term solutions	“How can we solve this together?”
Mediation	If third-party needed	Ask instructor or peer leader

Practice 4

Conflict Analysis

Scenario

Your team has two dominant speakers. Two others never contribute. You are the facilitator.

Group Tasks

❖ Identify the type of conflict.

❖ Analyze the root cause.

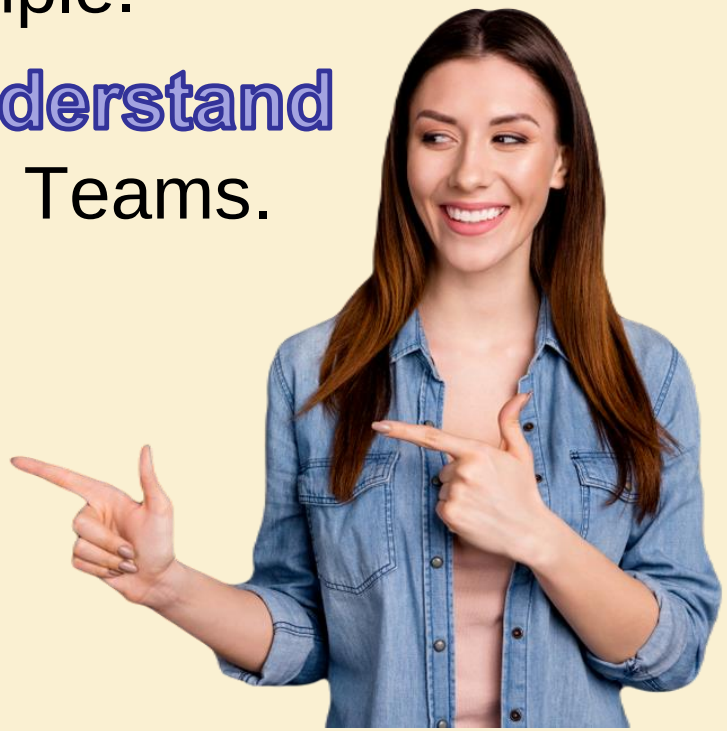
➡ Write your solution and resolution strategy (to be presented and shared).

- ***Your tasks:***
  - ❖ 5 - 10 members per group.
  - ❖ 15 minutes preparation.
  - ❖ Present your solution and resolution strategy.
  - ❖ The classmates vote on clarity, logic, and delivery.

In conclusion

- What is a Problem?
- Structured Problem-Solving Frameworks?
- The 5 Whys Technique – Example.
- Brainstorming Tips.
- Common Sources of Conflict in Teams.
- Conflict Escalation Model.
- Conflict Resolution Strategies.

Understand



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THANK YOU
FOR YOUR ATTENTION

Q&A